SIMATS ENGINEERING

CSA17 – ARTIFICIAL INTELLIGENCE LAB

Day 1 – Question 5: 5. Missionaries and Cannibals Problem

Aim: To transport three missionaries and three cannibals safely across a river in a boat carrying one or two people.

Python Program

```
from collections import deque

def valid(m, c):
    return (0 <= m <= 3 and 0 <= c <= 3 and
            (m == 0 or m >= c) and
            (m == 3 or 3-m >= 3-c))

start = (3, 3, 1)
goal = (0, 0, 0)
moves = [(1, 0), (2, 0), (0, 1), (0, 2), (1, 1)]
q = deque([(start, [start])])
visited = {start}

while q:
    state, path = q.popleft()
    if state == goal:
        for s in path:
            print(s)
        break

    m, c, boat = state
    for dm, dc in moves:
        if boat == 1:
            new = (m-dm, c-dc, 0)
        else:
            new = (m+dm, c+dc, 1)

        if valid(new[0], new[1]) and new not in visited:
            visited.add(new)
            q.append((new, path + [new]))
```

Sample Output

```
Output: A valid sequence of states is printed.
Each tuple is (missionaries on left, cannibals on left, boat side).
Boat side 1 = left bank; 0 = right bank.
```

Result: The program demonstrates the stated Artificial Intelligence problem.